

RELIANCE JIO: LEVERAGING LATE MOVER ADVANTAGE¹

Dr. V. S. Pai wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

This publication may not be transmitted, photocopied, digitized, or otherwise reproduced in any form or by any means without the permission of the copyright holder. Reproduction of this material is not covered under authorization by any reproduction rights organization. To order copies or request permission to reproduce materials, contact Ivey Publishing, Ivey Business School, Western University, London, Ontario, Canada, N6G 0N1; (t) 519.661.3208; (e) cases@ivey.ca; www.iveycases.com. Our goal is to publish materials of the highest quality; submit any errata to publishcases@ivey.ca.

Copyright © 2021, Ivey Business School Foundation

Version: 2021-04-28

In July 2020, Mukesh Ambani, the chief executive officer (CEO) of Reliance Industries Ltd. (RIL), was reflecting on the progress made by his company. He knew that although the company had been extremely successful in the past several decades, its future lay elsewhere, since both oil refining and petrochemicals, RIL's flagship businesses, were based on fossil fuels, an exhaustible resource. His ambition, coupled with hard work and leveraging RIL's fledgling telecommunications (telecom) business, had catapulted RIL to becoming India's most valuable company and among the top 50 companies globally.² To achieve this distinction, RIL had recently made several acquisitions worth nearly US\$2 billion³ in media, telecom, and digital technology; these included several start-ups such as chat box platform Haptik, online music subscription company Saavn, and e-fashion destination Fynd. These acquisitions represented a departure from RIL's past practices, when most developments had happened in-house via greenfield projects. This change in its growth strategy was part of a plan to ruthlessly reshape the company's growth model in line with global realities, full of disruption, while side-tracking linear and static growth models.

The intent was to create an entire ecosystem that allowed people to experience digital life to the fullest. This was to be executed by placing all the digital services—online retail, entertainment (including content streaming), digital payments, education, and health care—on the Jio telecom network.⁴ Reliance Jio focused on the telecom service business and together with Jio Platforms, a holding company, managed other digital businesses of RIL. Thus, it was to provide a one-stop access to a host of utilities. The logic driving this thinking was that the market valued aggregation-type, technology-driven businesses more highly compared with traditional (commodity) businesses under the premise that they drove efficiency.

Ambani needed to ensure that the gamble would pay off. He took a long time from the initial investment in the telecom sector in 2010 to the actual commercial launch of his business six years later. He chose to enter the market with a new generation of technology, while ignoring the existing technologies. He pursued massive investments, large-scale, and all-India operations with a low-price strategy. Would this approach pay off? What else would he need to do to manage in an industry in which several technologies were converging? His investments were well positioned to be leveraged to enter several other businesses; however, doing so would increase risk and competition.

EVOLUTION OF THE TELECOM MARKET

Since the mid-1990s, the Indian telecom sector had undergone monumental transformation aided by policy and regulatory shifts. India's National Telecom Policy of 1994 and The New Telecom Policy of 1999 set the ball rolling to usher in substantial change in the sector. The Department of Telecommunications (DoT)⁵ was India's only service provider in the early 1990s and therefore played a dominant role. Because of significant reforms introduced in 1994, India's telecom service shifted from a monopoly to a competitive market.

The country was segregated into 22 telecom circles, and the DoT allocated telecom licences to telecom players via auctions at the circle level. State-owned Bharat Sanchar Nigam Ltd. (BSNL)⁶ grew substantially by reducing tariffs and improving efficiency.⁷ As competition intensified, the government intervened by establishing, through an act of Parliament, a sectoral regulator, the Telecom Regulatory Authority of India (TRAI). For speedy settlement of disputes, The Telecommunications Dispute Settlement and Appellate Tribunal was set up.

Soon, mobile services outgrew the fixed telephone service. The former had started in 1997 and from the start had been dominated by private players. The government ensured that no single firm monopolized the market by following a policy of managed competition, ensuring that at least four or five firms in each circle received licences. Private-sector operators adopted two of the then latest technologies, namely global system for mobile communications (GSM) and code division multiple access (CDMA).⁸ The Indian government sold licenses to the mobile phone spectrum in India via two rounds of auction in 1998 and 2004. In 2008, the government decided to auction the spectrum on a first-come, first-served basis, which turned out to be controversial, as auctioning resulted in the spectrum being captured by a few operators at the expense of the rest. The latter approach to the sale of spectrum was alleged by some to be an unholy nexus between some government officials and a few telecom companies, and this finally resulted in court cases. Ultimately, the spectrum licences had to be cancelled, leading to losses and a lot of bad feelings.⁹

The industry then saw a shakeout, with smaller players exiting and some large players buying them out via mergers and acquisitions, leading to consolidation. As a result of these activities, consumers benefited from increased quality, a wider variety of services, and falling prices. The role of the government went through major changes. It started off as a market participant, shifted to the role of a facilitator, and ultimately became a regulator. As the dust began to settle, disruptive technologies and the massive scale of operations led to a more concentrated market. As a consequence of the mergers and acquisitions, only three major private players and one government-owned firm remained.¹⁰

The industry had seen phenomenal growth since the mid-1990s. Starting from a minuscule market of 14.5 million connections in 1997, India became the second-largest telecom market in the world with over 1.19 billion subscribers at the end of September 2019 (see Exhibit 1).¹¹ Private operators outnumbered public sector players, with the former capturing 88.81 per cent of the market while the latter managed the remaining 11.19 per cent.¹² India's teledensity (i.e., telephone density, or the number of landline connections per 100 individuals in a given area) stood at 90.52, with urban areas having 160.63 and rural areas, 57.59.¹³ Besides better connectivity, the quality of service and the affordability of telecom services grew by leaps and bounds. India's Internet subscriber base also rose substantially to more than 650 million with more than 10 Internet service providers serving the market (see Exhibit 2).¹⁴

RIL'S DYNAMICS OF GROWTH

Reliance Industries Ltd.¹⁵ was set up in 1966 by Shri Dhirubhai H. Ambani as a small textile manufacturing unit. Over the years, the company transformed itself into a large conglomerate. Its operations included hydrocarbon exploration and production, petroleum refining and petrochemical marketing, textiles, retail,

and telecommunications. In March 2020, RIL was India's largest private-sector company on most financial measures. It had several firsts to its credit. In 1996–97, it was the first Asian firm to issue bonds in the United States. In 2004, it became "the first Indian private sector organization to be listed on the Fortune Global 500 list."¹⁶ By the end of March 2019, RIL's annual sales stood at ₹6,228.09 billion,¹⁷ with net profit of ₹395.88 billion.¹⁸

The company entered the organized retail business in 2006 through its subsidiary, Reliance Retail. In 2018–19, Reliance Retail's annual revenues reached ₹1,305.56 billion, which was the highest annual revenue by any Indian retail company. Its profit before depreciation, interest, and taxes (PBDIT) for the same year stood at ₹19.23 billion.¹⁹ During this year, it extended its retail reach to more than 6,600 cities and operated 10,415 stores covering an area of 2 million square metres (22 million square feet). In June 2010, RIL ventured into the telecom space by acquiring a substantial stake in Infotel Broadband Services (P) Ltd. and secured entry into all telecom circles in the country. However, it was only in 2016 that its wholly owned subsidiary, Reliance Jio Infocomm Ltd. (R-Jio), commenced services with the Jio Welcome Offer. At the time of R-Jio's entry, the telecom market was experiencing intense competition, with approximately 10 other players (see Exhibit 3). Within six months of launching its operations, R-Jio had acquired a subscriber base of more than 100 million.²⁰

R-JIO'S ENTRY STRATEGY

On September 5, 2016, R-Jio commenced its wireless telecommunication services across India. The thrust of its strategy was rock-bottom tariff plans for voice calls, speed, and data. At that time, more than 20,000 tariff plans were being offered by existing operators. R-Jio tried to simplify its offering and make it affordable to consumers. Voice calls were ₹0.49 per minute, and its average 1 gigabyte (GB) data rate was ₹205. Within a few months of R-Jio's launch, these rates fell drastically, to ₹0.19 and ₹19, respectively. Customers took note of these developments and responded positively, leading to a marked increase in R-Jio's subscriber base. The competitors, however, experienced an extremely difficult and challenging situation.²¹

To begin with, R-Jio had introduced an offer to provide free unlimited services for customers who bought its service within 90 days of launch. Under this welcome offer, subscribers were entitled to unlimited Long-Term Evolution (LTE) data,²² and national voice, video, and messaging services, including national roaming, all completely free up until December 31, 2016. Then, having observed some loophole in the existing government regulation, R-Jio extended the offer for a further period of 90 days, to the joy of would-be subscribers. The company had set a target of acquiring 100 million subscribers during this network capacity testing period, which it achieved comfortably, resulting in approximately 8 per cent of the market share. According to Trushar Barot of NiemanLab, "In the year since it launched, Jio has acquired over 100 million users—many connecting to the mobile internet for the first time in their lives. To give some context, this effectively makes Jio the fastest adopted technology in human history."²³

The groundwork for entering the telecom services market had started in 2010 when RIL, the parent company, acquired 95 per cent equity in Infotel Broadband Services (P) Ltd. (Infotel) for \$1 billion. At that time, Infotel was the only firm that possessed broadband spectrum in all of India's 22 telecom zones, which, unsurprisingly, was one of the critical reasons why RIL decided to buy it out.²⁴ The company wanted to be proactive and lay the foundation for its future strategy. Even before commencing its commercial operations, R-Jio had laid 250,000 kilometres of fibre-optic cable with the intent of establishing network coverage in 18,000 urban centres and approximately 200,000 rural areas. Further, RIL's agreement with Anil Dhirubhai Ambani Group (ADAG)²⁵ included a non-compete clause that was nullified via a truce between the warring brothers, Mukesh and Anil Ambani.²⁶

As part of its entry strategy, R-Jio decided to offer only 4G service. Thus, unlike its competitors, it was not burdened with maintaining and operating 2G and 3G infrastructure, which were both expensive and moving toward obsolescence. Being a greenfield player, R-Jio invested in a telecom network that was capable of not just servicing 4G but could easily be upgraded to 5G, 6G, and future spectrums.²⁷

To complement its technological dominance, R-Jio attempted to disrupt the market further by investing in a massive scale of operations to drive down costs. Scale also helped when bargaining with back-end companies—including, among others, handset manufacturers, tower companies, and agencies that laid fibre-optic cable—to reduce its costs to a fraction of what its competitors had incurred. To enhance its competitive advantage, R-Jio created a comprehensive product for customers, coordinating inexpensive hardware, advanced technology at attractive prices, and a galaxy of software offerings. The offerings included unlimited free voice calls, JioTV with high-definition (HD) streaming on the phone, low-priced 4G phones under its LYF brand,²⁸ and a Wi-Fi router with access to its network to enable data usage on multiple devices.²⁹

Being a late mover in the market, R-Jio could employ the latest technology to offer superior services. It was also not burdened with redundant technology. On the contrary, it was keen to be future-ready by investing in the emerging 5G technology, which its competitors found difficult to do because of their burden of past borrowings.³⁰ With deep pockets and the intent to become market leader, R-Jio offered services for free or at low prices. Thus, R-Jio turned on its head the theory of early-mover advantage, taking the entire industry by a hurricane of prudent strategic business decisions and calculated financial risks. This approach was, of course, not without its price—Jio had made an initial investment of \$25 billion.³¹ Neil Shah, an analyst at a market research firm, commented on Jio's entry into the telecom sector: "It was a huge risk and they really had to prove themselves. . . . But the vision was right, the timing was right, and the scale was massive."³²

Skirmishes between incumbents and R-Jio had begun well before the formal launch of its operations. R-Jio pointed fingers at rivals, stating that they were trying to sabotage the test phase of the launch. It alleged that its subscribers experienced a larger number of dropped calls since competitors refused to make available a sufficient number of interconnection points with their networks. Rivals made counterclaims, saying that R-Jio's own systems had shortcomings that resulted in the dropped calls. Analysts agreed, noting that the back-up system to manage voice calls did not exist with R-Jio, which had a mobile network run entirely on 4G technology.³³

In December 2016, when R-Jio announced that the launch policy of freebies would continue for another three months, the existing players were further incensed. According to Sunil Bharti Mittal, the CEO of Bharti Airtel, India's second-largest telecom player based on number of subscribers, "Having such a long, free promotional period and in some sense, decided by laws of the land in their favour, is unheard of. In my opinion, in Europe or US, this would have been stopped. It would have been seen as predatory."³⁴

Unfortunately, the incumbents had underestimated not only Jio but also the telecom customers. Existing competitors were dismissive of 4G, emphasizing India's low adoption of high-speed data technology. They pronounced that most subscribers would be unwilling to pay for what was being offered, as the price would be prohibitive. However, what happened was the opposite. By making it affordable, R-Jio proved that subscribers loved fast data and forced competitors to roll out their own 4G networks.³⁵

Questions were raised regarding the profitability of RIL's huge investment in telecom. But surprisingly, in January 2018, R-Jio reported its maiden net profit for the quarter October–December 2017, just over a year after commencing operations. Even here it was alleged that the company had capitalized a large proportion of its costs and therefore showed a \$78 million profit. Otherwise, it would have reported a net loss of \$376 million for the period.³⁶

THE IMPACT OF R-JIO'S ENTRY

After the entry of R-Jio, several fundamental changes were observed. For instance, data usage per smart phone saw a steep rise to 9.8 GB per month. This trend was expected to increase further with the passage of time. Tariffs fell sharply, which increased data usage.³⁷ By increasing its monthly data consumption to 1 billion GB of data, up from 200 million GB, India became the world's number-one country in mobile data use—within barely six months of R-Jio's launch.³⁸ R-Jio network data revealed that, each month, each subscriber consumed, on average, high levels of digital usage: 10 GB of data, 700 minutes of voice, and 134 hours of video. Also, voice over long-term evolution (VoLTE)³⁹ had also grown substantially,⁴⁰ which could play an important role in modernization and technological progress.

The fall in data prices induced new segments of society to use and benefit from this technology. Telecom customers from the bottom of the pyramid, who, in the past, could not afford data services, rushed to subscribe⁴¹ to R-Jio's offerings. The digital applications offered were intended to cater to consumer needs spread across various sectors, including education, health, entertainment, and banking. In the process of diffusing telecom technology to the masses, the company recorded a customer acquisition cost of less than \$1 per customer, substantially lower than the existing industry norm of \$25.⁴²

R-Jio's entry into the market brought several other dramatic and far-reaching developments. It changed the face of India's telecom industry by shifting the basis of competition from voice to data. The upheaval in the telecom service market led to India's largest merger, between Vodafone and Idea; cornered the market leader, Bharati Airtel, into posting losses after more than 15 years of profitable operations; and ensured that five telecom companies either shut their operations or were acquired by bigger competitors. The most significant beneficiaries were the customers.⁴³

Competitors were badly hit. Within the first two quarters of R-Jio's launch, the total revenue of its competitors dropped by 7 per cent. Bharti Airtel saw the biggest fall, of 72 per cent, in its net income in the quarter ending March 2017, and its profits declined by 4.9 per cent.⁴⁴ Both Vodafone and Idea, the second- and third-largest telecom firms before their merger, experienced revenue declines of 8.3 per cent and 19 per cent, respectively⁴⁵. R-Jio's rivals had no choice but to reduce their tariffs, to match those of R-Jio, resulting in a sharp decline in industry profits, which did not show any signs of abating. The scale of disruption began to emerge when in late 2016 Vodafone wrote down the value of its Indian business⁴⁶ by €5 billion.⁴⁷ However, competitor Airtel grabbed the opportunity to gobble up laggards Tata Teleservices Ltd. and Norway's Telenor. Reliance Communication thought that it would tide over the crisis and stay afloat by making a last-ditch attempt to merge with Malaysia's Airtel. But the deal fell through, and Anil Ambani decided to sell Reliance Communication to brother Mukesh.⁴⁸

By March 2019, Reliance Jio Infocomm Ltd. (R-Jio) was among India's top three telecom players in terms of subscriber base (see Exhibit 4) and by revenue market share (see Exhibit 5). But fierce competition continued between the late entrant R-Jio, the well-entrenched Bharti Airtel, and the merged entity Vodafone-Idea. Their market shares stood at 31.75 per cent, 30 per cent, and 28 per cent respectively in December 2019. By March 2020, R-Jio had increased its subscriber base to 34 per cent,⁴⁹ with many of those gains at the cost of Vodafone-Idea Ltd. R-Jio thus became India's number-one telecom operator in terms of both subscriber base and revenue market share.

Of R-Jio's three competitors, Vodafone seemed the most negatively impacted. After entering the Indian telecom market, it faced a series of major challenges. The first shock came to Vodafone over its acquisition of Hutchison Whampoa's stake in the latter's Indian operations. The Indian government believed that it should receive a share in the profits of this transaction, for which it enacted a retrospective tax policy. Somehow the crisis blew over without affecting the company. The next debacle Vodafone faced was the

advent of R-Jio. Loaded with debt owing to expensive payments made during previous spectrum auctions, Vodafone's cost structure rose threateningly. It had yet to recover from this situation when the final hit arrived, as the regulator slashed interconnection usage charges (IUCs)⁵⁰ from ₹0.14 to ₹0.06 per minute. Vodafone, which had enjoyed a large number of incoming calls, was badly hit. R-Jio, on the contrary, benefited, as a result of the reduced costs of delivering voice calls to rival networks.⁵¹

Vodafone, however, was hopeful that its investments in 2G and 3G networks that enabled it to reach the poorest of the poor, especially in remote areas, would assist it to flourish. This market was out of reach of R-Jio, as it offered handset and services based on 4G technology.⁵²

Frustrated, Vodafone hit out at the regulator after it slashed IUCs. Its group CEO remarked, "We only ask for a level-playing field in terms of regulation. I think it's fair to say that for the last two years, we have had many regulatory outcomes that have worked against everyone in the market except Jio. We have made these points clear even earlier."⁵³ R-Jio dismissed this view and wrote to the government conveying its perspective:

We further submit that the failure of two operators, even in the unlikely event of it actually happening, will not have an impact on the sector dynamics with existence of vibrant competition including presence of the PSUs [Public Sector Undertakings] and there is no restriction on entry by new operators. Further, there will be no impact on digitization and government programmes, as these operators, anyway were not investing sufficiently in the sector and have been claiming a financial stress for a long time now and they have not shown any inclination to modernize the networks, as evident for Trai [Telecom Regulatory Authority of India] data in its IUC consultation paper.⁵⁴

On August 31, 2018, Vodafone and Idea merged to create India's biggest telecom company by sales revenue and subscriber base. In March 2017, their combined sales revenue and earnings before interest, taxes, depreciation, and amortization (EBITDA) stood at ₹8,159.20 billion and ₹2,440.10 billion, respectively. In March 2020, the merged entity's annual revenue and EBITDA had dropped to ₹4,497.80 billion and ₹521.00 billion, respectively. The fall of EBITDA was so severe that it was inadequate to cover even its interest costs. Further, during the three-and-a-half-year period, the annual revenue per user (ARPU) had reduced from approximately ₹180 to ₹109.⁵⁵

Analysts at a leading equity research firm observed that, to a large extent, Vodafone–Idea should have been held responsible for its own decline. In their assessment, the company had played it safe by adopting a conventional approach to simply hang on, expecting better times to arrive. They also pointed out that the company had misread the behaviour of subscribers: "We are particularly disappointed with the pace of capex [capital expenditure] deployment by VIL (Vodafone Idea Ltd). LTE network lags both R-Jio and Bharti on coverage as well as capacity and time is of utmost essence as VIL can ill afford to let the gap expand further."⁵⁶

Several factors were responsible for Vodafone–Idea's status, including an unstable regulatory environment, high operational costs, hyper-intensive competition, and very low tariffs. Bharti Airtel's CEO, Sunil Mittal, opined that consolidation in the industry had happened in a haphazard manner. As a consequence, investments worth approximately \$50 billion had been lost. Industry observers felt that Vodafone had seen its ups and downs in the Indian market but its demise, if it were to happen, would lead to much disruption in the sector and a future of foreign direct investment (FDI) in India, as Vodafone's investment was one of India's largest FDIs.⁵⁷

THE ROAD AHEAD

Mukesh Ambani, always ahead of the pack, immediately displayed a keenness to embrace 5G technology when the government announced the possibility of allocating spectrum toward the end of 2019. This government announcement came within six months of R-Jio having struck a deal with Ericsson, Nokia, and

Huawei to provide 5G network service. The company wanted to commence its own 5G trials based on open standards and interoperability. These efforts were intended to reduce dependence on Samsung, the Korean vendor, which was the sole provider of 4G network solutions to R-Jio.⁵⁸

In a statement, Ambani outlined his vision of creating a digital society for every Indian by leveraging R-Jio:

To drive the next leg of growth, a truly transformational and disruptive digital services company has been set up, which will bring together India's number one connectivity platform, leading digital app ecosystem and world's best tech capabilities, for creating a truly digital society for each Indian.⁵⁹

After having disrupted the mobile telephone market, R-Jio's focus shifted to shaking up the fixed broadband market with the launch of Jio GigaFiber service, a fibre-to-the-home (FTTH) service that was ultrafast and Wi-Fi-enabled. The company was in the process of offering Internet speeds of between 100 megabits per second (Mbps) and 1 gigabit per second (Gbps), which could enable the masses to connect to its digital infrastructure. Although R-Jio offered this service at a fraction of the global rates, with subscription starting at ₹700, the customer response was lukewarm, likely because the cost of FTTH service to both homes and businesses was more expensive than mobile services. The FTTH service also required wiring individual homes, and because the process was time-consuming, many homes remained to be connected. Over the next few years, the company's ARPU was expected rise on account of FTTH service, resulting in R-Jio becoming the largest contributor to RIL's earnings.⁶⁰

To make Jio GigaFiber attractive to subscribers, its welcome offer included free voice calls for life from landline phones, a free high-definition light-emitting diode (LED) TV, a dish antenna, and a set-top box. The digital set-top box offered ultra-high-definition entertainment, virtual reality content, multi-party video conferencing, voice-enabled virtual assistants, interactive gaming, home security, and several smart-home solutions. Regarding Jio Fiber, Ambani said, "The Jio GigaFiber service will come bundled with subscriptions to most leading premium OTT⁶¹ applications," and then continued: Also, for the first time in India, we are introducing a disruptive concept for watching new movies. Premium Jio Fiber customers will be able to watch movies in their living rooms the same day these movies are released in theatres! We call this Jio First-Day-First-Show."⁶²

R-Jio started projecting itself not just as a telecom company but as an Internet-of-Things (IoT)⁶³ solutions entity and indicated that this focus would be its critical growth engine in the days to come, as part of its Digital India dream. Ambani saw a ₹200 billion opportunity in the narrowband IoT market. To take advantage of this opportunity, he started work to create a pan-India network of connected devices. To be able to create digital solutions, R-Jio hired several thousands of engineers and invested in approximately a dozen start-ups.⁶⁴

Specifically, R-Jio introduced on its all-India 4G network an innovative service, Narrowband Internet-of-Things, or NB-IoT. However, the company faced competition in this area from both Vodafone-Idea and Airtel. Vodafone-Idea had already rolled out this service in Kochi, Jaipur, Bangalore, and Chennai; and Airtel was also on the verge of introducing the same.⁶⁵

Rumours suggested that Vodafone-Idea might shut down under the weight of the Supreme Court's decision that it repay the government adjusted gross revenues (AGR)⁶⁶ of approximately ₹500 billion. Speculation was rife in the media that R-Jio and Airtel were in consultation on the possible need to quickly upgrade their networks to absorb the former's subscribers. Many in the media debated which of the remaining two players would benefit more if Vodafone-Idea shuttered. Some industry observers believed that, since the majority of Vodafone-Idea users belonged to 2G and 3G categories, they would flock to Airtel and even to the state-owned BSNL, since R-Jio offered only the 4G network.⁶⁷ Others argued that, in the absence of

Vodafone–Idea, Airtel would lose the cost advantage of a shared tower infrastructure, which would reduce its cost structure by up to 15 per cent in terms of network and customer servicing costs and lead to higher tariffs, resulting in uncompetitive pricing. It would also mean a trusted partner would be lost, thereby weakening Airtel's abilities to manage in a challenging regulatory environment and to face a strong adversary such as R-Jio.⁶⁸

According to some experts, the Indian government was not inclined to disturb the three large competitors' existing market structure, as doing so could have several undesirable consequences for consumers, industry, and the country, including not being able to realize the government's goal of achieving \$5 trillion in gross domestic product by 2024. Restricting the existing value chain would also negatively impact the country's Digital India dream.⁶⁹ Customers in rural and remote areas, served by 2G and 3G networks, would be the worst hit as the remaining players might not be able to handle the extra subscriber load while maintaining a high quality of service.

A major development in April 2020 was Facebook's acquisition of a 9.99 per cent equity stake in Jio Platforms with an investment of \$5.7 billion, or ₹435.74 billion.⁷⁰ Although R-Jio was a late entrant, its valuation shot up because of its dominant position in telecom and its linkages with the retail and digital spaces, which offered immense future possibilities. Jio Platforms was the fully owned subsidiary of RIL and contained the digital services of the group. R-Jio was a wholly owned subsidiary of Jio Platforms, which also owned content-generation ventures and other of RIL's digital businesses, including MyJio, JioTV, JioCinema, JioNews, and JioSaavn. R-Jio, via this partnership, accessed Facebook's technological expertise and its talent pool and reduced the company's debt burden. Facebook's CEO, Mark Zuckerberg, wrote of the deal on his Facebook page: "The country is in the middle of a major digital transformation, and organizations like Jio have played a big part in getting hundreds of millions of Indian people and small businesses online."⁷¹

Along with this partnership, Reliance Retail (a group company) and WhatsApp (the messaging app owned by Facebook) entered into a tie-up to strengthen the former's e-commerce business to be spearheaded by R-Jio's small business initiative, JioMart. JioMart Platform would strengthen its support services to small businesses, including the nearest small merchants on WhatsApp. Consequently, Reliance Retail would be able to, along with its physical footprint, penetrate rural markets via digital commerce using WhatsApp. A rough estimate indicated that this partnership would empower approximately 30 million small stores to digitally transact with customers in their neighbourhood.⁷² Competition would be intense in the electronic payments space that the R-Jio–Facebook partnership had ventured into. In anticipation of the global competition the partners said, "Facebook and [RIL have] . . . made it clear that the first order of business is establishing a cashless system to anchor forays into Internet commerce mobile services. That alliance inserts a powerful new competitor into an arena already contested by Google, Walmart, Amazon and . . . Paytm."⁷³

Mukesh Ambani, who was from the fossil fuels business, was attracted to the overcrowded, technologically disruptive, and heavily regulated telecom industry. Despite being a late entrant, R-Jio was able to become the undisputed market leader within a few years, while well-entrenched incumbents struggled to stay afloat. The future looked challenging as the telecom sector continued to be impacted by dynamically changing technologies. Investments in the telecom sector were also linked to Reliance Retail and could galvanize the group's retail business. Digital India was a challenging yet attractive commercial proposition with significant societal outcomes for R-Jio. Ambani's R-Jio needed to ensure that its massive investments and the risks it had taken led to the expected outcomes.

**EXHIBIT 1: SNAPSHOT OF INDIA'S TELECOM SUBSCRIBERS AS OF THE QUARTER ENDING
SEPTEMBER 30, 2019**

Telecom Subscribers (Wireless + Wireline)	
Total Subscribers	1,195.24 million
Urban Subscribers	677.95 million
Rural Subscribers	517.29 million
Market Share of Private Operators	88.81%
Market Share of Public Operators	11.19%
Total Teledensity	90.52
Urban Teledensity	160.63
Rural Teledensity	57.59
Wireless Subscribers	
Total Subscribers	1,173.75 million
Urban Subscribers	659.18 million
Rural Subscribers	514.56 million
Market Share of Private Operators	89.75 %
Market Share of Public Operators	10.26%
Total Teledensity	88.90
Urban Teledensity	156.18
Rural Teledensity	57.28

Note: Teledensity = telephone density.

Source: Created by the author using information from Telecom Regulatory Authority of India, *The Indian Telecom Services Performance Indicators, July-September 2019*, January 8, 2020, accessed March 26, 2020, https://main.trai.gov.in/sites/default/files/PIR_08012020_0.pdf.

**EXHIBIT 2: INDIA'S INTERNET SERVICE PROVIDERS AND SUBSCRIBERS
AS OF THE QUARTER ENDING SEPTEMBER 30, 2019**

Rank	Service Provider	Number of Subscribers	Share %
1	Reliance Jio Infocomm Ltd.	355,926,487	51.76
2	Bharti Airtel Ltd.	154,880,425	22.52
3	Vodafone Idea Ltd.	140,342,199	20.41
4	BSNL	28,216,649	4.10
5	Atria Convergence Tech Pvt. Ltd.	1,479,233	0.22
6	MTNL	1,164,440	0.17
7	Hathway Cable & Datacom Pvt. Ltd.	861,946	0.13
8	You Broadband Pvt. Ltd.	773,597	0.11
9	GTPL Broadband Pvt. Ltd.	301,750	0.04
10	Excitel Broadband Pvt. Ltd.	294,000	0.04
	Total of Top 10 Internet Service Providers	684,240,726	99.51
	Others	3,382,700	0.49
	Grand Total	687,623,426	100

Source: Created by the author using information from Telecom Regulatory Authority of India, *The Indian Telecom Services Performance Indicators, July-September 2019*, January 8, 2020, accessed March 26, 2020, https://main.trai.gov.in/sites/default/files/PIR_08012020_0.pdf.

EXHIBIT 3: MARKET SHARE OF INDIA'S MOBILE TELECOM COMPANIES IN JUNE 2016, BEFORE THE ENTRY OF RELIANCE JIO INFOCOMM

Service Provider	Market Share
Bharti Airtel	24.71%
Vodafone	19.26%
Idea Cellular	17.03%
Reliance Communication	9.54%
BSNL/MTNL	9.00%
Aircel	8.59%
Others	11.57%
Total	100

Source: Created by the author using information from Telecom Regulatory Authority of India, *The Indian Telecom Services Performance Indicators, April–June, 2016*, 12, December 1, 2016, accessed May 13, 2020, https://traf.gov.in/sites/default/files/Indicator_Reports_April_June_01_12_2016.pdf.

EXHIBIT 4: SUBSCRIBER BASE OF INDIA'S MAJOR WIRELESS TELECOMS COMPANIES, 2014–15 TO 2018–19 (IN MILLIONS)

Service Provider	2013–14	2014–15	2015–16	2016–17	2017–18	2018–19	% Growth over FY 2017–18
Bharti Airtel	205.39	226.02	251.24	273.65	304.19	325.18	20.99
Vodafone	166.56	183.80	197.95	209.06	222.70	394.84	(39.07)
Idea	135.79	157.81	175.07	195.37	211.21		
R Com	110.89	109.47	102.41	83.50	0.19	0.02	(0.17)
BSNL	94.65	77.22	86.35	100.99	111.68	115.74	4.06
Aircel	70.15	81.40	87.09	90.90	74.15	—	(74.15)
R-Jio	—	—	—	108.68	186.56	306.72	120.16

Note: FY = financial year; R Com = Reliance Communications; R-Jio = Reliance Jio Infocomm Ltd.; Vodafone and Idea merged in 2018.

Source: Created by the author using information from Telecom Regulatory Authority of India, *Annual Report 2018-19*, November 2019, accessed March 26, 2020, https://main.traf.gov.in/sites/default/files/Annual_Report_15012020_0.pdf.

EXHIBIT 5: SALES AND PROFITS COMPARISON OF INDIA'S MAJOR TELECOM COMPETITORS, 2017–18 AND 2018–19 (IN ₹ MILLIONS)

Company	2017–18		2018–19	
	Sales	Profit	Sales	Profit
Vodafone–Idea	282,420	(4,168)	369,860	(146,040)
Bharti Airtel	536,630	792	496,080	(18,190)
R-Jio	201,540	723	388,380	29,640

Note: R-Jio = Reliance Jio Infocomm Ltd.

Source: Compiled from “Annual Reports | Vodafone Idea Limited,” Vodafone Idea Limited, n.d., accessed March 12, 2020, www.vodafoneidea.com/investors/annual-reports; “Quarterly and Annual Results,” Airtel India, n.d., accessed March 12, 2020, www.airtel.in/about-bharti/equity/results; “Financial Reporting,” Reliance Jio, n.d., accessed March 12, 2020, www.ril.com/InvestorRelations/FinancialReporting.aspx.

ENDNOTES

- ¹ This case has been written on the basis of published sources only. Consequently, the interpretation and perspectives presented in this case are not necessarily those of Reliance Industries Ltd. or any of its subsidiaries or employees.
- ² Priyabrata Prusty, "Mukesh Ambani's RIL among 50 Most Valued Companies in the World as M-cap Tops \$173 Billion," *Times Now News*, July 23, 2020, accessed December 2, 2020, www.timesnownews.com/business-economy/markets/article/mukesh-ambanis-ril-among-50-most-valued-companies-in-the-world-as-shares-rise-over-18-in-a-month/625993.
- ³ All currency amounts are in US\$ unless otherwise indicated.
- ⁴ Reliance Industries Limited, *Integrated Annual Report 2018–19*, 2, 2019, accessed December 1, 2020, www.ril.com/DownloadFiles/CorporateAnnouncements/AR_14981_RELIANCE_2018_2019_19072019205801.pdf.
- ⁵ The Department of Telecommunications (DoT) was a department of the Ministry of Communications in the Government of India.
- ⁶ Bharat Sanchar Nigam Ltd. (BSNL) was one of India's largest and leading public sector units, providing a comprehensive range of telecom services in India.
- ⁷ Prashant Pandhre, "The Evolution of India's Telecom Industry: From Monopoly to duopoly to a competitive structure," August 13, 2017, accessed April 3, 2020, <https://nationalinterest.in/telecom-industry-in-india-from-monopoly-to-duopoly-to-a-competitive-structure-2d92383814b2>.
- ⁸ Cellular service had two main competing network technologies: Global System for Mobile Communications (GSM) and Code Division Multiple Access (CDMA). CDMA was a form of multiplexing that allowed numerous signals to occupy a single transmission channel, optimizing the use of available bandwidth. GSM used a variation of time division multiple access (TDMA). It digitized and compressed data, then sent it down a channel with two other streams of user data, each in its own time slot.
- ⁹ Pandhre, op. cit.
- ¹⁰ Ibid.
- ¹¹ Created by the author using information from Telecom Regulatory Authority of India, *The Indian Telecom Services Performance Indicators, July–September 2019*, January 8, 2020, accessed March 26, 2020.
- ¹² Ibid.
- ¹³ Ibid.
- ¹⁴ Ibid.
- ¹⁵ "The Reliance Story," Reliance Industries Ltd, accessed December 2, 2020, www.ril.com/TheRelianceStory.aspx; "Reliance Industries Ltd., (Reliance) – Company History," *Business Standard*, accessed December 2, 2020, www.business-standard.com/company/reliance-industr-476/information/company-history.
- ¹⁶ Ibid.
- ¹⁷ ₹ = INR = Indian rupee; ₹1 = US\$0.0144 on March 31, 2019.
- ¹⁸ Reliance India Ltd Annual Report 2018-19, www.ril.com/ar2018-19/index.html.
- ¹⁹ Reliance India Ltd Annual Report 2018-19, www.ril.com/ar2018-19/ril-annual-report-2019.pdf.
- ²⁰ Rasul Bailay and Varun Jain, "Reliance Retail crosses Rs 100,000 crore revenue mark," *The Economic Times*, April 18, 2019, accessed March 29, 2020, <https://retail.economictimes.indiatimes.com/news/industry/reliance-retail-crosses-rs-100000-crore-revenue-mark/68943752>; Aveek Datta, Lison Joseph and Baiju Kalesh, "RIL enters telecom," *Mint*, June 12, 2010, accessed March 25, 2020, www.livemint.com/Home-Page/RlFdW825DrfDobiamHeO5J/RIL-enters-telecom.html; Telecom Regulatory Authority of India, *The Indian Telecom Services Performance Indicators*, April–June, 2016, December 1, 2016, accessed May 13, 2020, https://tra.gov.in/sites/default/files/Indicator_Reports_April_June_01_12_2016.pdf.
- ²¹ Aditya Narayan, "Disrupting the telecom industry – Reliance Jio: Entry Strategy," Jan 23, 2019, accessed March 16, 2020, www.linkedin.com/pulse/disrupting-telecom-industry-reliance-jio-market-entry-narayan.
- ²² Long-Term Evolution (LTE) was a standard for wireless broadband communication for mobile devices and data terminals and was based on the GSM/EDGE (Global System for Mobile Communications/Enhanced Data for Global Evolution) technologies; 4G (or fourth generation) was a common name given to LTE technology—i.e., 4G and LTE were synonymous.
- ²³ Amil Kapoor and Chirag Yadav, *Impact of Reliance's Entry: A Socio-Economic Analysis of Jio-fication and India's GDP Story*, Institute for Competitiveness, March 12, 2018, accessed March 27, 2020, <https://competitiveness.in/impact-of-reliance-entry/>.
- ²⁴ Pratish Narayanan, "Reliance to buy Infotel for \$1 billion," June 11, 2010, accessed March 15, 2020, www.livemint.com/Home-Page/d7lx4R54LoQtmMpNzmJ1MM/Reliance-to-buy-Infotel-for-1-bn.html.
- ²⁵ Reliance Anil Dhirubhai Ambani Group was an Indian conglomerate headquartered in Mumbai. The company, formed after Dhirubhai Ambani's business was divided, was headed by his younger son, Anil Ambani.
- ²⁶ Yogesh Sapkale, "Jio: Why it is different," Oct 03, 2016, accessed March 15, 2020, www.moneylife.in/article/jio-why-its-different/48301.html.
- ²⁷ Aditya Narayan. op. cit.
- ²⁸ LYF smart phones launched in India in 2016 and ran on Google's Android operating system. They come preloaded with Reliance Jio apps, including MyJio, Jio Express News, JioChat, JioNet Hotspot, and others.
- ²⁹ Aditya Narayan. op. cit.
- ³⁰ Ibid.
- ³¹ Ibid.
- ³² Simon Mundy, "Reliance Industries' Jio Reshapes India's Telecoms Market," *Financial Times*, March 15, 2018, accessed March 27, 2020, www.ft.com/content/3f1fe4d6-e4e0-11e7-a685-5634466a6915.
- ³³ "Reliance Jio Call Drops Due to Its Under-preparedness: Airtel," *Economic Times*, September 28, 2016, accessed November 30, 2020, https://economictimes.indiatimes.com/news/economy/policy/reliance-jio-call-drops-due-to-its-under-preparedness-airtel/articleshow/54543166.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst.

- ³⁴ Anuj Srivas, "How Reliance Jio's Entry Tied Regulatory Knots around India's Telecom Ecosystem," *The Wire*, January 13, 2018, accessed April 26, 2020, <https://thewire.in/tech/reliance-jio-telecom-regulation-trai-anil-ambani>.
- ³⁵ Mundy, op. cit.
- ³⁶ Ibid.
- ³⁷ "Jio's Entry Led to \$10 Billion Annual Savings for India: Report," April 6, 2018, accessed March 16, 2020, <https://economictimes.indiatimes.com/tech/internet/jios-entry-led-to-10-billion-annual-savings-for-india-report/articleshow/63637786.cms>
- ³⁸ Prerana Bhat, "The Entry of Reliance Jio and the Resultant Tremors on the Indian Telecom Markt," January 3, 2020, accessed March 27, 2020, www.indianfolk.com/entry-reliance-jio-resultant-tremors-indian-telecom-market/
- ³⁹ The big advantage of VoLTE was that call quality was superior to 3G or 2G connections, as far more data could be transferred over 4G than over 2G or 3G.
- ⁴⁰ Institute for Competitiveness, "The Impact of Reliance's Entry", accessed March 27, 2020, <https://competitiveness.in/impact-of-reliance-entry/>.
- ⁴¹ Ibid.
- ⁴² Ibid.
- ⁴³ Navadha Pandey, "Why competition in telecom is key to the digital India dream," November 14, 2019, accessed April 2, 2020 https://images.livemint.com/img/2019/11/14/original/gggdgdgd_1573752612957.png; Kiran Rathee, "Reliance Jio effect: Bharti Airtel posts Rs 9.4-bn loss in India operations," July 27, 2018, accessed April 2, 2020, www.business-standard.com/article/companies/reliance-jio-effect-bharti-airtel-posts-rs-9-4-bn-loss-in-india-operations-118072700072_1.html.
- ⁴⁴ Prerana Bhat. op. cit.
- ⁴⁵ Ibid.
- ⁴⁶ Mundy. op. cit.
- ⁴⁷ € = European euro; €1 = US\$0.9508 on December 31, 2016.
- ⁴⁸ Mundy. op. cit.
- ⁴⁹ "Reliance Jio May Gain 10% Telecom Market Share to Reach 44% by FY22: Report," *Business Standard*, March 6, 2020, accessed July 11, 2020, www.business-standard.com/article/companies/reliance-jio-may-gain-10-telecom-market-share-to-reach-44-by-fy22-report-120030600824_1.html.
- ⁵⁰ Interconnect usage charge (IUC) was a charge the telecom service provider of a caller pays to the telco on whose network the call terminated. It was related to 2G services and outdated as it was not relevant to advanced generation technologies.
- ⁵¹ Navadha Pandey. op. cit.
- ⁵² Ibid.
- ⁵³ Ibid.
- ⁵⁴ Ibid.
- ⁵⁵ Deborshi Chaki and Mobis Philipose, "Inside the Battle to Save Vodafone Idea," *Mint*, March 3, 2020, accessed April 12, 2020, www.livemint.com/industry/telecom/inside-the-battle-to-save-vodafone-idea-11583165208038.html.
- ⁵⁶ Chaki and Philipose. op. cit.
- ⁵⁷ Romit Guha and Gulveen Aulakh, "Telcos Wrote Off up to \$50 Billion Due to Reliance Jio: Sunil Mittal," *Economic Times*, November 27, 2017, accessed April 3, 2020, https://economictimes.indiatimes.com/news/company/corporate-trends/telcos-wrote-off-up-to-50-billion-due-to-reliance-jio-sunil-mittal/articleshow/61811188.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst.
- ⁵⁸ Sohini Mitter, "Reliance Jio to Start Own 5G Trials, Seeks Government Nod," *YourStory*, March 3, 2020, accessed March 31, 2020, https://yourstory.com/2020/03/reliance-jio-5g-trials-government-nod-mukesh-ambani?utm_pageloadtype=scroll.
- ⁵⁹ Sohini Mitter, "Inside Jio's billion-dollar deals: Why investors are flocking to Mukesh Ambani's \$65B internet biz", *Your Story*, May 28, 2020, accessed March 12, 2020, <https://yourstory.com/2020/05/jio-billion-dollar-deals-investors-mukesh-ambani-internet-biz>.
- ⁶⁰ Surender T, "How Mukesh Ambani is Rewriting RIL's future," February 6, 2020, accessed April 3, 2020, www.fortuneindia.com/enterprise/reimagining-reliance-industries-the-mukesh-ambani-way/104110; "Reliance Jio GigaFiber vs Airtel vs ACT Fibernet: Broadband Speed, Data Plans, Price Comparison," August 27, 2019, accessed April 3, 2020, www.businesstoday.in/technology/news/reliance-jio-gigafiber-vs-airtel-vs-act-fibernet-broadband-speed-data-plans-price-comparison/story/372004.html.
- ⁶¹ OTT referred to over-the-top content, when audio, video, and other media content was delivered over the Internet without the involvement of a multiple-system operator.
- ⁶² "Reliance Jio's 'First-Day-First-Show' Service under Its GigaFiber Plan: Things to Know," *Times of India*, August 12, 2019, accessed January 17, 2020, <https://timesofindia.indiatimes.com/business/india-business/reliance-jios-first-day-first-show-service-under-its-gigafiber-plan-things-to-know/articleshow/70643098.cms>.
- ⁶³ IoT was a network of Internet-enabled devices, all having an Internet protocol address and communicating with each other through a mobile application (app) on a smart phone interface.
- ⁶⁴ Navadha Pandey, "All Eyes on Jio's Connected Car Tech as It Gears to Offer a Glimpse of Future," *Livemint*, February 5, 2020, accessed March 30, 2020, www.livemint.com/auto-news/all-eyes-on-jio-s-connected-car-tech-as-it-gears-to-offer-a-glimpse-of-future-11580867381104.html.
- ⁶⁵ Danish Khan, "Reliance Jio to Connect Billion Connected Devices via NB-IoT; eyes Rs 20K-cr Opportunity Per Year," August 12, 2019, accessed April 12, 2020, <https://telecom.economictimes.indiatimes.com/news/jio-to-connect-billion-connected-devices-via-nb-iot-eyes-rs-20k-cr-opportunity-per-year/70640854>

⁶⁶ Some telecom firms were required to pay adjusted gross revenues (AGR) to the Indian government. Since the dues ran into several hundreds of billions of rupees, paying them could jeopardize the existence of the telecom firms, thereby affecting competition, customer service, and government revenues in future.

⁶⁷ Navadha Pandey, "Jio, Airtel Brace for the Worst, Plan Upgrades in Case Voda Idea Fails," Mint, February 18, 2020, accessed March 31, 2020, www.livemint.com/industry/telecom/jio-airtel-brace-for-the-worst-plan-upgrades-in-case-voda-idea-fails-11582043068964.html.

⁶⁸ Kalyan Parbat, "Bharti Airtel Faces High Costs, Isolation If Vodafone Idea Exits," *Economic Times*, February 16, 2020, accessed March 31, 2020, <https://economictimes.indiatimes.com/industry/telecom/telecom-news/bharti-airtel-faces-high-costs-isolation-if-vodafone-idea-exits/articleshow/74144218.cms?from=mdr>.

⁶⁹ Deborshi Chaki and Mobis Philipose, "Inside the Battle to Save Vodafone Idea," Mint, March 3, 2020, accessed March 31, 2020, www.livemint.com/industry/telecom/inside-the-battle-to-save-vodafone-idea-11583165208038.html.

⁷⁰ Mike Isaac and Vindu Goel, "Facebook Invests \$5.7 Billion in Indian Internet Giant Jio," *New York Times*, April 21, 2020, accessed May 14, 2020, www.nytimes.com/2020/04/21/technology/facebook-jio-india.html.

⁷¹ Ibid.

⁷² "Jio-Facebook Deal Looks at Local Level e-Commerce: Mukesh Ambani," Mint, April 22, 2020, June 25, 2020, www.livemint.com/companies/people/jio-facebook-deal-looks-at-local-level-e-commerce-mukesh-ambani-11587536876933.html.

⁷³ P R Sanjai and Saritha Rai, "Visa Stake Caps \$8 Billion Deal Run for Reliance," Bloomberg, May 8, 2020, accessed March 12, 2020, www.bloomberg.com/news/articles/2020-05-08/visa-investment-caps-8-billion-deal-run-for-asia-s-richest-man.